

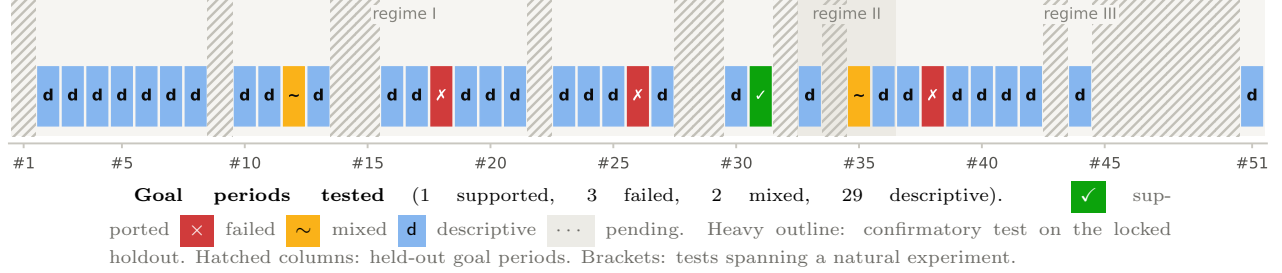
H65: Leaders are routers, not sources

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

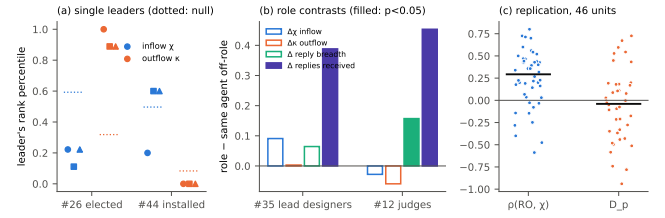
Bottom line. Designated leaders are attention hubs, not content routers; the one elected leader is a content source.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test asks whether designated leaders aggregate (high content inflow, broad replies, low outflow) rather than broadcast. Inflow χ and outflow κ come from a read-gated linear-response fit on context-ledger reads, with an unread in-flight term for co-response. Reply rates come from DQ2. The fit runs on 46 period units and four native tests: an elected leader (#26), rotating lead designers (#35), an installed leader (#44) and rotating judges (#12). Synthetic runs came first and recalibrated the single-leader test. **No leader is a router** (0/4 natives). Lead designers and judges get +0.39 and +0.45 more replies per statement ($p \leq 0.007$), and judges answer more agents. Their content inflow does not rise. The elected leader is a content *source*. Across units, answering raises inflow ($\rho +0.29$, 36/46), but attention is unrelated to content flow. **Mixed.** The holdout is not run.



Observables. (a) Single leaders: rank percentiles of inflow (blue) and outflow (orange) in three embedding variants; dotted lines are the skeleton-null means. The #26 leader is a source. (b) Within-agent role contrasts: the role adds replies received (and reply breadth for judges), not content inflow or outflow. (c) Replication: answering correlates with inflow; attention alignment D_p centres on 0.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not install a leader to collect or relay information. In this village the role buys attention: about 0.4 more replies per statement and, for judges, broader answering. It does not make the leader absorb more content or pass it on. An elected leader can become a content source, which an operator should treat as an agenda field on the swarm. To route information, address the agents who must act on it; a hub does not relay it. Per-agent content coefficients are noisy, and same-room messages the agent had not yet read carry 0.6 of the read response.

PHYSICAL MODEL

Directed linear response (model 02) and information flow (model 04). The content state z_B of agent j 's statement is a whitened 32-d embedding with the goal and kickoff field projected out. It responds to j 's own past, the day $\times$ room field f , read human messages H , unread in-flight messages U , and the decayed read input S_{iB} from each agent i . Inflow χ_j is the slope on all reads. Outflow κ_i is the extra slope on i 's reads in other agents' statements.

$$z_B = az_P + a'e_j + bf + hH_B + \lambda U_B + \sum_i \kappa_{i \rightarrow j} S_{iB} + \xi_B$$

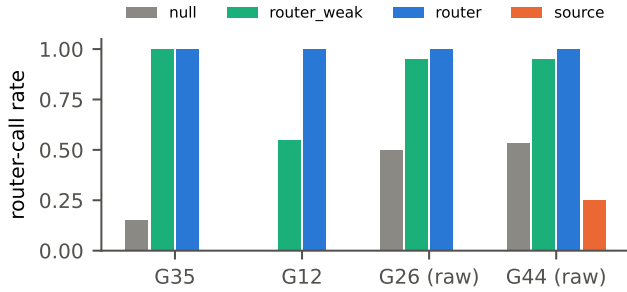
$$S_{iB} = \frac{\sum_{A \in i, \text{ read}} w_A z_A}{1 + \sum w_A}, \quad w_A = e^{-\text{age}/15 \text{ min}}$$

$$\chi_j : \kappa_{i \rightarrow j} \equiv \chi_j \quad \forall i, \quad \rho_R = \text{pct}(\chi) - \text{pct}(\kappa)$$

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RESULTS VS PREDICTIONS

Prediction	Observed (bge; gte; style)	Verdict
S3 null router calls at base rate	0–0.15 within agent; 0.50 single leader	recalibrated
R1 answering is absorbing	$\rho + 0.29$ (36/46); +0.35; +0.28	yes
R2 attention goes to sinks	$D_p - 0.04$ (17/40); -0.07 ; -0.06	no
R3 inflow $\neq$ outflow	$\rho(\chi, \kappa)$ 0.14; 0.09; 0.14	yes
R4 agrees with H32 outflow	median ρ 0.43 (27/32)	yes
N1 #26 elected router	pct χ 0.22, pct κ 1.00 (null 0.32)	no: source
N2 #35 lead designers	$\Delta\chi + 0.09$ (p 0.18; gte 0.009); $\Delta\kappa$ 0; replies +0.39	mixed
N3 #44 installed	27 statements; no call possible	descr.
N4 #12 judges	$\Delta\kappa - 0.06$; $\Delta\chi - 0.03$; replies +0.45, breadth +0.16	mixed



Second figure. Synthetic router-call rates on the real skeletons (20 replicates per cell, 60 for single-leader nulls). Within-agent designs (#35, #12) recover planted routers and stay quiet under the null. The raw single-leader percentile rule fires in half of null replicates, because the skeleton biases one agent’s rank. Those natives were recalibrated on their own null before the real run.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 $\sim$ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (writeup/paper.tex).

CAVEATS

Per-agent coefficients are noisy at role-window scale; the #35 inflow contrast is significant under one embedding model only. Each single-leader native has one leader, so it is a weak test. Inflow includes answering mechanics, because DQ2 parents are partly content-selected. Unread in-flight messages carry 0.6 of the read response. The #44 leader has 27 statements.

ROUND 2

- **Essence.** whether any agent in a swarm relays information (a two-hop path through it), and what makes an elected leader a content source.
- **H65-R1** A direct relay test.
- **H65-R2** The agenda event.
- **H65-R3** Co-response-matched inflow.
- **H65-R4** Run confirm.py after committing (#45, #28).

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